

A
Report on
Inter Disciplinary Mini Project

**“A DATA-DRIVEN RESEARCH AND ANALYTICS
PLATFORM FOR GREEN SYNTHESIS OF
SEMICONDUCTOR MATERIALS”**

SUBMITTED TO THE PUNYASHLOK AHILYADEVI HOLKAR SOLAPUR
UNIVERSITY, SOLAPUR IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF

**BACHELOR OF TECHNOLOGY DEGREE IN
COMPUTER SCIENCE AND ENGINEERING**

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UNDER THE GUIDANCE OF
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N. B. NAVALE SINHGAD COLLEGE OF ENGINEERING,
SOLAPUR - 413 255
2026-27

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SOLAPUR



CERTIFICATE

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is a bonafide work carried out by above students under the supervision of Prof. S. S. Shakhapure and it is submitted towards the partial fulfillment of the requirement of Punyashlok Ahilyadevi Holkar Solapur University, Solapur for the award of the degree of Bachelor in Technology (Computer Science and Engineering) during academic year 2026-27.

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DECLARATION

We hereby declare that the work embodied in this Project Report Entitled “**A Data-Driven Research and Analytics Platform for Green Synthesis of Semiconductor Materials**” is carried out by us in partial fulfillment of the degree Bachelor in Technology (Computer Science and Engineering) from N. B. Navale Sinhgad College of Engineering, Solapur and we have not submitted the same to any other University/Institute for the award of any other degree. The work presented in this report is original and has been carried out under the supervision of Prof. S. S. Shakhpure, Department of Computer Science and Engineering. All sources of information and references have been duly acknowledged.

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ACKNOWLEDGEMENT

We feel profound happiness in forwarding this project report as an image of our sincere efforts put into the design and development of the **GreenSynth Analytics Platform** — a data-driven research and analytics system for the green synthesis of semiconductor materials.

Our sincere thanks to our respected Project Guide, **Prof. S. S. Shakhapure**, Department of Computer Science and Engineering, N. B. Navale Sinhgad College of Engineering, Solapur, for his invaluable guidance, constant encouragement, and constructive criticism throughout all phases of this project. His technical insight and research-oriented thinking have been instrumental in shaping the scientific and engineering quality of this work.

We extend our heartfelt gratitude to our Project Coordinator, **Prof. V. V. Palmur**, for his meticulous coordination of the project review process and for consistently maintaining the high academic standards expected of this interdisciplinary work.

We express our deep respect and gratitude to **Prof. Harish T. Gurme**, Head of the Department of Computer Science and Engineering, for providing an environment of intellectual freedom and access to resources that made the development of a system of this technical scale possible within an academic setting.

We are equally thankful to our respected Principal, **Dr. S. D. Nawale**, for his vision of excellence and for fostering a culture of research and innovation at our institution.

We sincerely thank all the faculty members of the Department of CSE for their academic support, constructive feedback during project reviews, and for creating an atmosphere that encourages interdisciplinary thinking and rigorous engineering practice.

Finally, we express our gratitude to our families and peers for their unwavering moral support, patience, and encouragement throughout the duration of this project.

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ABSTRACT

The synthesis of nanocrystalline semiconductor thin films through green chemistry routes presents significant ecological and economic advantages over conventional vacuum-based fabrication techniques. However, the realization of this potential is severely constrained by multi-parametric process complexity, fragmented and non-traceable data management practices, and the absence of a closed-loop computational optimization framework suited to laboratory-scale experimental research.

This report presents **GreenSynth Analytics** — a full-stack, data-driven research and analytics platform developed specifically for the systematic study, characterization, analysis, and optimization of the green synthesis of semiconductor materials. The primary experimental campaign (Project P7) targets the phytochemical-mediated synthesis of nanocrystalline Copper Oxide (CuO) thin films using Mulberry (*Morus*) leaf extract as a natural reducing and capping agent, deposited onto heated substrates by Ultrasonic Pneumatic Spray Pyrolysis.

The platform provides an end-to-end scientific workflow covering: Design of Experiments (DOE) generation using Full Factorial, Fractional Factorial, Central Composite Design (CCD), and Box-Behnken methods; management of synthesis parameters across thirteen controllable process variables; multi-technique instrumental characterization analysis encompassing X-Ray Diffraction (XRD) with Scherrer crystallite size determination, UV-Vis spectroscopy with Tauc plot band-gap extraction, Fourier-Transform Infrared Spectroscopy (FTIR), Scanning Electron Microscopy (SEM), and DC Electrical I-V characterization; statistical comparison using Pearson and Spearman correlation, Ordinary Least Squares regression, and one-way ANOVA; supervised machine learning model training with k-fold cross-validation across five regression architectures; experimental prediction validation with signed error analysis and model drift detection; evidence-based experiment candidate recommendation; and automated scientific PDF report generation.

The system is implemented as a three-tier architecture using FastAPI and PostgreSQL for the backend and React/TypeScript for the frontend, with SHA-256 cryptographic checksum verification enforcing raw data immutability. A synthetic dataset of 400 experiments with 1,380 characterization runs was used to benchmark the complete pipeline. The champion machine learning model (Random Forest) achieved a cross-validated R^2 of 0.6685 and an RMSE of 0.1501 S/cm for prediction of electrical conductivity from synthesis parameters alone.

GreenSynth demonstrates that a rigorously engineered, scientifically honest research software platform can substantially compress the experimental iteration cycle for green semiconductor synthesis, providing a reproducible, traceable, and computationally augmented environment for evidence-based materials discovery.

Keywords: Green Synthesis, Copper Oxide Thin Films, Spray Pyrolysis, Machine Learning, Design of Experiments, Research Data Management, Scientific Analytics, Interdisciplinary Platform, FastAPI, React, PostgreSQL.

Project Report Outline

The report is organized chronologically in terms of the objectives accomplished.

- **Chapter One** introduces the project, defines the problem statement, outlines the scope and objectives, identifies the need, and establishes the motivation and relevance to an interdisciplinary approach.
- **Chapter Two** presents a literature review covering existing data management and analytics systems for materials synthesis, identifies limitations of current approaches, defines the research gap, and introduces the proposed system and SDLC model.
- **Chapter Three** specifies functional and non-functional requirements, hardware and software requirements, and the system environment.
- **Chapter Four** describes the system architecture, analysis models, UML diagrams (Use Case, Class, Sequence), ER diagram, data flow diagrams, and the data dictionary.
- **Chapter Five** details the technologies used, scientific algorithms, database design, module descriptions, and key implementation screenshots.
- **Chapter Six** presents the testing strategy, unit and integration test cases, and performance evaluation results.
- **Chapter Seven** discusses system outputs, ML model benchmarking results, comparison with existing systems, and industrial benefits.
- **Chapter Eight** summarizes the work accomplished, achievements, industrial usefulness, and future enhancement directions.
- **Appendix A** lists abbreviations used throughout the report.
- **Appendix B** lists publications associated with this work.
- **References** provide the complete bibliography.

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Chapter 1

INTRODUCTION

The global transition toward sustainable and high-performance semiconductor technologies has created an urgent demand for synthesis processes that are simultaneously cost-effective, environmentally responsible, and scientifically rigorous. The field of green synthesis — which replaces hazardous chemical reducing agents with natural plant extracts — has emerged as a promising route toward environmentally compatible fabrication of metal oxide semiconductor nanostructures. However, the practical realization of this potential is severely limited by the complexity of the multi-parametric synthesis process, the inadequacy of existing data management practices in academic laboratories, and the absence of a principled computational framework for experimental optimization.

GreenSynth Analytics is a full-stack, data-driven research and analytics platform designed to address these limitations. It provides a scientifically rigorous, end-to-end computational environment for the design, execution, analysis, and optimization of green semiconductor synthesis experiments. The platform targets the phytochemical-mediated synthesis of nanocrystalline Copper Oxide (CuO) thin films — a p-type metal oxide semiconductor with demonstrated utility in photovoltaics, photocatalysis, and gas sensing — using Mulberry (*Morus*) leaf polyphenol extract as a natural capping agent, deposited via Ultrasonic Pneumatic Spray Pyrolysis (USP).

1.1 Introduction to Project

GreenSynth Analytics is a full-stack research software platform developed to connect laboratory experimental data with scientific characterization analysis, statistical comparison, machine learning-based property prediction, and evidence-based experimental optimization. It is engineered as a three-tier application: a Python/FastAPI asynchronous REST API backend, a PostgreSQL relational database with cryptographic data integrity enforcement, and a React/TypeScript frontend with a responsive, scientific-grade user interface.

The primary experimental campaign (Project P7) focuses on:

- **Target Material:** Nanocrystalline Copper Oxide (CuO) — monoclinic phase, p-type semiconductor
- **Green Capping Agent:** Mulberry (*Morus*) leaf phytochemical extract (polyphenols, flavonoids) dissolved in ethanol
- **Deposition Technique:** Ultrasonic Pneumatic Spray Pyrolysis (USP) onto glass, FTO glass, ITO glass, quartz, and silicon substrates
- **Characterization Suite:** X-Ray Diffraction (XRD), UV-Visible Spectroscopy (UV-Vis), Fourier-Transform Infrared Spectroscopy (FTIR), Scanning Electron Microscopy (SEM), and DC Electrical I-V measurement

- **Primary Optimization Target:** Electrical Conductivity (σ , S/cm)

The system encompasses the complete experimental lifecycle. It begins with Design of Experiments generation, proceeds through synthesis parameter management and raw file ingestion, continues with physics-based characterization analysis (Scherrer crystallite size, Tauc band-gap, Ohmic resistivity), advances through statistical comparison and machine learning model training, and culminates in evidence-based experimental recommendation and automated PDF report generation. Every calculated result is cryptographically traceable to its raw source file, analytical algorithm, and version, making scientific irreproducibility architecturally impossible within the platform.

The system supports up to eight concurrent research projects sharing a common data schema and analytical infrastructure. It implements five core scientific integrity principles: (1) complete data traceability, (2) strict categorical isolation between MEASURED, CALCULATED, STATISTICAL, PREDICTED, and VALIDATED data, (3) immutability of raw experimental files, (4) configuration-driven project definitions, and (5) scientifically honest output — no unvalidated ML predictions are exposed to the user without explicit uncertainty quantification and applicability domain confirmation.

1.2 Problem Statement

The synthesis of semiconductor thin films through green chemistry routes offers significant ecological and economic advantages. However, the practical realization of this potential is constrained by a series of unresolved scientific and operational challenges:

1. **Multi-Parametric Process Complexity:** Spray pyrolysis of phytochemically stabilized metal oxide precursors involves no fewer than thirteen controllable synthesis parameters: substrate temperature, precursor concentration, mulberry extract concentration and volume, ethanol volume, spray rate, spray duration, nozzle-to-substrate distance, carrier gas pressure, number of spray cycles, ambient temperature, and ambient relative humidity. The response surface governing electrical conductivity, crystallite size, and optical band gap is fundamentally non-linear, exhibiting sharp thresholds (e.g., a crystallization onset near 230°C, an optimal phytochemical passivation window at 15–20 g/L extract concentration), multi-factor interaction effects, and parameter-dependent regime shifts. Manual, one-factor-at-a-time (OFAT) experimental strategies are statistically inadequate to characterize this response surface and will systematically fail to locate global optima.
2. **Fragmented and Non-Traceable Data Workflows:** In typical academic laboratory practice, raw instrumental data (XRD patterns, UV-Vis spectra, FTIR transmittance curves, I-V sweeps) are stored in ad hoc spreadsheets or personal directories with no formal provenance, no version control of calculated results, and no cryptographic integrity verification. Accidental overwriting of raw data, unversioned recalculation with updated formulae, and undetected transcription errors are endemic — directly threatening the scientific reproducibility of published results.
3. **Absence of a Closed-Loop Optimization Framework:** Current practice treats experimental design, laboratory execution, data analysis, and subsequent

experiment planning as disconnected activities performed in sequence without formal computational support. There exists no integrated system that can generate statistically rigorous DOE matrices, ingest and validate characterization data, train and validate physics-aware ML predictive models, and generate ranked, constrained experiment candidates for the next iteration — all within a unified, scientifically honest software environment.

4. **Premature or Unsupported Machine Learning Application:** The growing enthusiasm for applying machine learning to materials science data has produced models trained on small, heterogeneous, or improperly cleaned datasets, without applicability domain checks, target leakage detection, or rigorous cross-validation. Such predictions appear numerically precise but carry no reliable physical validity, and in the absence of a rigorous validation layer, can misdirect expensive experimental resources.
5. **Reproducibility Crisis in Green Synthesis Research:** Published green synthesis results for CuO thin films exhibit wide variability in reported electrical conductivities (ranging from <0.01 S/cm to >1.0 S/cm under nominally similar conditions). This variability is attributable not only to genuine process sensitivity but also to inconsistent analytical protocols, undocumented parameter co-variations, and absence of statistical uncertainty quantification. A platform that enforces categorical isolation of data types and prevents mixing of measurement categories in downstream inference is essential to restore credibility to comparative analysis.

GreenSynth Analytics is developed as a direct, engineered response to each of these problems.

1.3 Scope and Objectives

1.3.1 Scope

The scope of this project is defined along three primary axes: scientific domain, platform capability, and data integrity boundaries.

Scientific Domain Scope: The platform is specifically scoped to the synthesis, characterization, and optimization of p-type metal oxide semiconductor thin films and nanoparticles produced via solution-based and phytochemically mediated green routes. Within this domain, the primary experimental campaign (Project P7) focuses on nanocrystalline CuO thin films synthesized by spray pyrolysis using Mulberry extract in ethanol. Characterization techniques in scope include XRD, UV-Vis spectroscopy, FTIR, SEM image management, and DC Electrical I-V analysis. Derived material properties include Scherrer crystallite size, Tauc plot optical band gap, electrical resistivity, and electrical conductivity.

Platform Capability Scope: The platform encompasses:

- Experimental project, experiment, and sample hierarchy management with full CRUD operations
- Synthesis parameter templates and version-controlled parameter sets with physical units

- Raw file immutability enforcement with SHA-256 cryptographic checksum verification
- Multi-technique characterization data ingestion, signal preprocessing, and scientific analysis
- Statistical multi-sample comparison (descriptive statistics, correlation analysis, OLS regression, ANOVA)
- Design of Experiments (DOE) generation: Full Factorial, Fractional Factorial, CCD, and Box-Behnken designs
- Machine learning dataset construction with target leakage detection and applicability domain enforcement
- Supervised regression model training (Linear Regression, Ridge, Random Forest, Gradient Boosting) with k-fold cross-validation
- Experimental prediction validation against actual laboratory outcomes with signed error analysis and model drift detection
- Evidence-based experiment candidate recommendation with objective scoring and constraint evaluation
- Automated scientific PDF report generation with full provenance traceability

Exclusions: The platform does not perform autonomous laboratory control or physical equipment interfacing. Deep SEM image segmentation, ab initio DFT calculations, and molecular dynamics simulation are outside the present scope. Synthetic/simulated data used for platform benchmarking is explicitly flagged and never presented as real empirical measurements.

1.3.2 Objective

Primary Objective: To design, implement, validate, and deploy a full-stack, scientifically rigorous research software platform that enables end-to-end data management, characterization analysis, statistical inference, machine learning modeling, and evidence-based optimization for the green synthesis of CuO semiconductor thin films via spray pyrolysis.

Secondary Objectives:

1. To establish a traceable, immutable experimental data architecture enforcing strict categorical isolation between raw measurements, derived calculations, statistical summaries, ML predictions, and experimentally validated results.
2. To implement and validate domain-specific scientific computation modules for XRD peak analysis, UV-Vis Tauc plot band-gap extraction, FTIR functional group identification, and DC electrical I-V Ohmic resistance fitting.
3. To develop a statistically rigorous multi-sample comparison engine capable of descriptive characterization, parametric and non-parametric correlation analysis, OLS regression, and one-way ANOVA.

4. To construct a machine learning pipeline with scientific integrity gating, including dataset eligibility auditing, target leakage prevention, applicability domain bounds enforcement, 5-fold cross-validation model selection, and prediction confidence interval estimation.
5. To implement a closed-loop experimental validation and retraining mechanism that computes signed, absolute, and relative prediction errors, detects model performance drift, and conditionally triggers model versioning.
6. To integrate a Design of Experiments engine supporting Full Factorial, Fractional Factorial, Central Composite, and Box-Behnken designs with randomized run ordering and direct conversion to planned laboratory experiments.
7. To build an evidence-based recommendation engine scoring synthesis parameter candidates against measurable objectives under physical domain constraints.
8. To produce automated, publication-quality scientific PDF reports with complete analytical provenance for all characterization results, statistical findings, ML model summaries, and optimization recommendations.

1.4 Identification of Need

The identification of need for GreenSynth Analytics arises from a convergence of emerging materials science priorities, unmet computational research infrastructure requirements, and critical shortcomings in the existing practice of green semiconductor synthesis.

1. Strategic Importance of CuO as a Semiconductor Material: Copper Oxide occupies a uniquely advantageous position among earth-abundant semiconductor materials. Its direct optical band gap of approximately 1.2–1.7 eV straddles the optimal solar energy absorption range; its p-type carrier transport properties make it a natural complement to n-type oxide layers in heterojunction photovoltaic structures; and its activity as a photocatalyst for the degradation of organic pollutants makes it relevant to environmental remediation. CuO is composed of copper — a relatively abundant, non-toxic transition metal — and oxygen, making it inherently compatible with green chemistry principles. The integration of phytochemical capping agents from Mulberry extract (polyphenols, flavonoids, quercetin, rutin, chlorogenic acid) further eliminates the use of synthetic reducing agents such as sodium borohydride (NaBH_4) or hydrazine, which are hazardous and environmentally persistent. Despite this promise, the full performance potential of phytochemically synthesized CuO thin films remains unrealized in the literature due to the absence of a systematic, statistically powered experimental campaign.

2. Inadequacy of Manual Data Management at Scale: A typical DOE-driven experimental campaign for spray pyrolyzed thin films generates per 400-experiment study: 6,000 synthesis parameter records, 1,380 raw instrumental characterization files, 1,881 derived property calculations, and numerous statistical and ML model artifacts. Manual management at this scale using spreadsheets is not merely inconvenient — it is scientifically unsafe. Errors in formula propagation, undetected row duplication, accidental overwriting of raw files, and absence of version-controlled calculated

results are structural failure modes of spreadsheet-based workflows. No freely available, domain-specific research platform exists that simultaneously provides immutable raw file management, physics-based characterization analysis, statistical inference, ML training with applicability domain gating, and DOE-driven optimization in a single scientifically honest system.

3. Demand for Reproducible and Statistically Valid Research: International scientific bodies have increasingly emphasized the reproducibility crisis as a systemic problem in experimental sciences. The lack of standardized computational workflows, documented analytical protocols, and transparent uncertainty quantification in materials synthesis research is a recognized root cause. A platform that architecturally enforces reproducibility through immutable raw data, version-controlled calculations, seeded DOE matrices, and cross-validated ML models directly addresses this need.

4. Industrial and Policy Relevance: Government initiatives including the European Green Deal, India’s National Green Hydrogen Mission, and the US Department of Energy’s Solar Futures Study have collectively underscored the urgency of accelerating the discovery and optimization of sustainable semiconductor materials for clean energy conversion. Computational tools that compress the experimental iteration cycle from months of trial-and-error to algorithmically guided, evidence-ranked experiment proposals directly serve this policy imperative.

1.5 Motivation

Scientific Motivation: Phytochemically synthesized CuO thin films prepared by spray pyrolysis sit at the intersection of several high-impact research frontiers: green chemistry, nanoscale semiconductor physics, thin-film photovoltaics, and heterogeneous photocatalysis. The electrical conductivity of these films is exquisitely sensitive to the interplay among substrate temperature (governing pyrolytic decomposition completeness and grain growth kinetics), phytochemical extract concentration (mediating grain boundary passivation at low concentrations but introducing insulating carbonaceous phase segregation at high concentrations), precursor concentration (controlling film stoichiometry and thickness), and spray dynamics (determining droplet evaporation uniformity and film continuity). This multi-factor, non-linear sensitivity makes the system an ideal candidate for systematic DOE-guided and ML-assisted optimization. The scientific question of what combination of green synthesis conditions yields maximum electrical conductivity in CuO thin films prepared from Mulberry extract remains open and unanswered in the published literature.

Methodological Motivation: The research community working on green synthesis of metal oxides has, to date, lacked a principled computational framework to move beyond OFAT experimentation. Published optimal synthesis conditions are local optima, not global ones, and are often not reproducible across laboratories due to undocumented parameter co-variations. Furthermore, ML applied to materials science data frequently suffers from target leakage, data snooping, and over-optimistic cross-validation, producing models that fail at inference on unseen conditions. The motivation to develop GreenSynth Analytics stems from the conviction that the scientific community deserves a platform that is transparently designed, architecturally honest, and that refuses to present a prediction as validated until laboratory evidence has confirmed it within a user-specified tolerance.

Engineering Motivation: The quality of scientific conclusions extracted from laboratory data is bounded above by the quality of the computational infrastructure used to process it. Building that infrastructure rigorously is therefore as much a contribution to science as performing the experiments themselves. This project demonstrates that software engineering discipline and materials science rigor are not merely complementary but deeply interdependent.

1.6 Relevance to Inter Disciplinary Approach

GreenSynth Analytics is inherently and irreducibly interdisciplinary. Its design, implementation, and scientific application simultaneously draw upon and contribute to six distinct academic and professional domains.

1. Materials Science and Solid-State Physics: The scientific foundation rests on solid-state materials physics: the Scherrer equation for XRD crystallite size analysis ($D = K\lambda/\beta \cos \theta$), the Tauc relation for optical band-gap determination from UV-Vis absorption spectra ($(\alpha h\nu)^2 \propto (h\nu - E_g)$ for direct transitions), Ohm's Law and resistivity theory for electrical transport characterization, and thermodynamic and kinetic models governing grain growth and phase purity in spray pyrolyzed metal oxides.

2. Chemistry — Green and Phytochemical Synthesis: The experimental domain is grounded in organic and inorganic chemistry: the phytochemical composition of Mulberry leaf extract (quercetin, rutin, chlorogenic acid, anthocyanins acting as electron donors and steric stabilizers), the stoichiometric decomposition of copper acetate precursors during pyrolysis, and the kinetics of CuO nucleation and grain coalescence on heated substrates.

3. Computer Science and Software Engineering: The platform constitutes a non-trivial software engineering artifact: an asynchronous REST API with SQLAlchemy ORM and Alembic database migrations, a React/TypeScript frontend with a custom design system, service-oriented architecture with strict separation of concerns, Docker-based containerized deployment, and a 118-test unit and integration test suite.

4. Statistics and Design of Experiments: The statistical and DOE modules draw directly from mathematical statistics: Pearson and Spearman correlation, OLS regression with significance testing, one-way ANOVA with F-statistic evaluation, and the formal theory of Full Factorial, Fractional Factorial, CCD, and Box-Behnken response surface designs.

5. Machine Learning and Artificial Intelligence: The ML pipeline applies supervised regression theory: bias-variance decomposition, k-fold cross-validation, ensemble methods (Random Forest, Gradient Boosting), regularization (Ridge regression), applicability domain theory for out-of-distribution detection, and prediction interval construction via residual variance estimation.

6. Data Engineering and Research Informatics: The platform operationalizes FAIR data principles (Findable, Accessible, Interoperable, Reusable): SHA-256 checksummed raw file immutability, Pydantic v2-validated metadata schemas, categorical provenance labeling, and atomic database transactions ensuring consistency across related records.

The distinctive contribution of this interdisciplinary approach is that no single discipline would have been sufficient: a materials scientist without software engineering

skills cannot build the platform; a software engineer without materials science knowledge cannot implement the Scherrer calculation correctly; a statistician without chemistry knowledge cannot structure a meaningful DOE for this system. The simultaneous application of all these domains makes GreenSynth Analytics a coherent, trustworthy, and genuinely interdisciplinary research tool.

Chapter 2

LITERATURE REVIEW

The literature relevant to GreenSynth Analytics spans four broad thematic areas: green synthesis of CuO semiconductor materials, research data management and laboratory informatics systems, machine learning applied to materials science, and Design of Experiments for process optimization. This chapter reviews key works in each area, identifies the limitations of existing approaches, defines the research gap, and introduces the proposed system within its SDLC model.

2.1 Existing Systems

Green Synthesis of CuO Nanostructures: Phytochemically mediated synthesis of copper oxide nanoparticles and thin films has attracted growing research interest over the past decade. Gawande et al. [1] provided a comprehensive review of copper-based nanomaterials for catalysis, establishing the scientific basis for CuO applications. Nasrollahzadeh et al. [2] demonstrated the use of plant extracts as green reducing agents for CuO nanoparticle synthesis. Specifically for Mulberry extract, several groups have reported synthesis of CuO nanoparticles and established that the polyphenolic and flavonoid content of *Morus* leaf extract effectively acts as a reducing and capping agent, limiting grain agglomeration and surface oxidation. Pillai et al. [3] reported Mulberry extract-mediated CuO nanoparticle synthesis with antibacterial activity. However, none of these studies employed systematic DOE or ML-assisted optimization; all used OFAT approaches with fewer than 30 experimental conditions, precluding reliable characterization of the multi-dimensional response surface.

Spray Pyrolysis for CuO Thin Film Deposition: Spray pyrolysis is a well-established scalable thin-film deposition technique. Moholkar et al. [4] reported spray pyrolyzed CuO films with electrical resistivities in the range of 0.5–50 Ω -cm depending on substrate temperature and precursor concentration. Thangaraju and Kaliannan [5] systematically varied substrate temperature from 200–450°C and reported an optimal crystallite size and electrical conductivity near 350–380°C for copper acetate precursors, consistent with the thermal activation profile modeled in GreenSynth. Serin et al. [6] investigated the effect of precursor concentration on the structural and electrical properties of spray pyrolyzed CuO films, reporting that optimal film continuity and conductivity required precursor concentrations between 0.15 and 0.30 M, consistent with the design space in Project P7.

Research Data Management and Laboratory Informatics: FAIR data principles (Findable, Accessible, Interoperable, Reusable) were formalized by Wilkinson et al. [7] and have become the de facto standard for scientific data management in computational materials science. Electronic Lab Notebook (ELN) systems such as LabArchives, Benchling, and the open-source eLabFTW provide general-purpose digital record-keeping for laboratory work. Platforms such as Materials Project [8] and AFLOW [9] provide curated databases of computationally derived materials properties (DFT) but are not designed for experimental raw data management, characterization

signal processing, or in-the-loop ML optimization. No existing open-source system simultaneously provides raw file immutability, domain-specific characterization analysis, closed-loop ML, and DOE integration for experimental green synthesis campaigns.

Machine Learning for Materials Discovery: Ward et al. [10] demonstrated the application of random forest regression to predict formation energies of binary compounds from compositional features, achieving cross-validated $R^2 > 0.90$ on DFT-computed datasets. Oliynyk et al. [11] applied gradient boosting classifiers to predict crystal structure types from elemental properties. For experimentally synthesized oxide films specifically, Pilania et al. [12] applied kernel ridge regression to predict band gaps of perovskite oxides, highlighting the importance of applicability domain bounds in preventing spurious extrapolations. Balachandran et al. [13] demonstrated active learning-guided experimental synthesis of piezoelectrics, demonstrating that the closed-loop ML paradigm can reduce the number of experiments needed to identify optimal compositions by 50–70% compared to OFAT strategies.

Design of Experiments in Materials Science: Montgomery [14] provides the definitive treatment of Design of Experiments methodology including Full Factorial, Fractional Factorial, Central Composite (CCD), and Box-Behnken designs. Mousounda et al. [15] applied CCD response surface methodology to optimize spray pyrolysis conditions for ZnO thin films, demonstrating significant improvement in film quality over OFAT approaches using only 29 designed experiments to characterize a 4-factor space. Khuri and Mukhopadhyay [16] reviewed RSM methodology with application to chemical and materials engineering, establishing the theoretical basis for the optimization engine in GreenSynth.

2.2 Limitations of Existing Systems

Based on the literature review, the following critical limitations of existing systems are identified:

1. **No Integration of Synthesis, Characterization, and Optimization:** Existing ELN systems (LabArchives, eLabFTW) record experimental notes but provide no domain-specific signal processing for characterization data (XRD peak fitting, Tauc plot extraction), no statistical comparison engine, no ML training pipeline, and no DOE generation. The researcher must manually export data to separate tools (OriginPro, MATLAB, Python notebooks) for each analysis step, with no provenance link between raw data and derived results.
2. **Absence of Raw Data Immutability:** None of the reviewed open-source ELN or LIMS (Laboratory Information Management System) platforms enforce cryptographic hash verification of raw instrumental files. A researcher can silently overwrite a raw XRD pattern after running a new experiment, breaking the traceability chain for all calculated properties derived from that file.
3. **No Applicability Domain Enforcement in Published ML Studies:** The majority of published ML studies for materials discovery report optimistic in-sample or poorly cross-validated performance metrics. Target leakage (including characterization properties as ML features when those properties are themselves derived from the synthesis conditions being optimized) is common. No published

study on green synthesis ML provides uncertainty-quantified predictions with applicability domain checks at inference time.

4. **OFAT Experimental Designs:** The overwhelming majority of green synthesis publications for CuO employ fewer than 20 experimental conditions, varying one parameter at a time. This strategy is incapable of detecting interaction effects between, for example, substrate temperature and extract concentration, which are physically expected and confirmed in the GreenSynth synthetic dataset.
5. **Non-Reproducible Analytical Protocols:** Published characterization results frequently do not specify the exact peak-fitting algorithm, baseline correction method, or Scherrer shape factor (K) used, making inter-laboratory comparison and result reproduction impossible without re-acquiring and re-analyzing the raw data.

2.3 Research Gap

Based on the limitations identified above, the following research gaps are established:

- **Gap 1:** No existing platform provides an integrated, end-to-end computational workflow for green synthesis experiments that connects DOE design, raw data ingestion with immutability enforcement, domain-specific characterization analysis, multi-technique statistical comparison, ML training and validation, and closed-loop experiment recommendation in a single scientifically honest system.
- **Gap 2:** No systematic, statistically powered DOE study has been published characterizing the multi-parameter response surface of electrical conductivity of CuO thin films synthesized by spray pyrolysis using Mulberry extract, across the full 13-dimensional synthesis parameter space.
- **Gap 3:** No ML model trained on experimental spray pyrolysis data for CuO or related metal oxide systems has been published with: (a) explicit target leakage prevention, (b) applicability domain bounds enforcement at inference time, (c) prediction uncertainty quantification, and (d) closed-loop validation against subsequent laboratory experiments.
- **Gap 4:** No open-source, domain-specific research informatics platform exists that operationalizes FAIR data principles for experimental green synthesis campaigns at the characterization data granularity (raw instrumental file + analysis run + calculated property), with SHA-256 integrity verification and categorical provenance labeling.

2.4 Proposed System

GreenSynth Analytics addresses all four identified research gaps with a unified, open-source, full-stack research software platform. The proposed system introduces the following novel contributions:

1. **Integrated End-to-End Scientific Workflow:** A single platform covering DOE → Experiment Management → Raw File Ingestion → Characterization Analysis → Statistical Comparison → ML Training → Prediction Validation → Optimization → Reporting.
2. **Immutable Raw Data Architecture:** All raw instrumental files (XRD, UV-Vis, FTIR, I-V CSV files) are stored with SHA-256 cryptographic checksums verified at ingestion and re-verified on access. Files may never be overwritten once finalized.
3. **Scientifically Honest ML Pipeline:** The ML module enforces target leakage prevention (upstream characterization properties are never included as synthesis-parameter-prediction features), applicability domain bounds checking at inference time, 5-fold cross-validation for model comparison, prediction uncertainty quantification via residual variance estimation, and a three-level experimental validation loop before any prediction is marked as validated.
4. **Evidence-Based Recommendation Engine:** Experiment candidates are scored against user-defined multi-objective functions (maximize conductivity, minimize crystallite size), evaluated against physical domain constraints, ranked, and presented for researcher approval before conversion to planned experiments.
5. **FAIR-Compliant Data Management:** Categorical provenance labels (MEASURED, CALCULATED, STATISTICAL, PREDICTED, VALIDATED) are applied at the record level and enforced in all query filters, ensuring data categories are never inadvertently mixed in downstream analysis.

2.5 SDLC Model

The GreenSynth Analytics platform was developed using an **Iterative and Incremental SDLC model**, organized into 21 sequential phases. Each phase delivered a vertically sliced, independently deployable and testable increment of functionality, with strict phase completion criteria: all unit tests pass, API endpoints return correct responses verified by integration tests, and database schema integrity is confirmed by Alembic migration validation.

This iterative approach allowed early-phase architectural decisions (database schema, API contract, data category model) to be validated against real scientific requirements before committing to downstream phase implementation, significantly reducing the cost of late-stage architectural changes.

Table 2.1: GreenSynth SDLC Phase Summary

Phase	Description	Status
1	Foundation & Architecture Setup	Completed
2	Project, Experiment & Sample Hierarchy	Completed
3	Synthesis Parameter Management & File Preservation	Completed
4	Scientific XRD Characterization & Preprocessing	Completed
5	UV-Vis Spectroscopy & Tauc Plot Band-Gap Analysis	Completed
6	Electrical Measurement & I-V Characterization	Completed
7	FTIR Spectroscopy & SEM Image Management	Completed
8	Sample Comparison & Statistical Analysis	Completed
9	Objective Definition & Design of Experiments	Completed
10	ML Dataset Preparation, Training & Prediction	Completed
11	Model Validation & Experimental Validation Loop	Completed
12	Recommendation Engine	Completed
13	Closed-Loop Learning & Autonomous Workflow	Completed
14	DOE Module Extension	Completed
15	Advanced Statistical Analysis & Evidence Layer	Completed
16	ML Prediction & Model Validation	Completed
17	Prediction Validation & Model Monitoring	Completed
18	Evidence-Based Optimization & Candidate Generation	Completed
20	Deployment, Production Hardening & Data Integrity	Completed
21	Scientific PDF Report Generation Module	Completed

Chapter 3

REQUIREMENT SPECIFICATION

3.1 Functional Requirements

The following functional requirements define the behavior that the GreenSynth Analytics platform must exhibit:

1. **FR-01: Project Management:** The system shall support creation, retrieval, update, and deletion of research projects. Each project shall maintain meta-data including project code (P1–P8), title, target material, synthesis technique, primary investigator, start date, and current status.
2. **FR-02: Experiment Management:** The system shall support creation, retrieval, update, and deletion of experiments under a parent project. Each experiment shall store a unique experiment code, synthesis technique, status (PLANNED, IN_PROGRESS, COMPLETED, FAILED), notes, and timestamps.
3. **FR-03: Sample Management:** The system shall support creation and management of physical samples associated with experiments. Each sample shall store substrate type, substrate dimensions, deposition date, and links to all characterization runs and calculated properties.
4. **FR-04: Synthesis Parameter Management:** The system shall store at least thirteen synthesis parameters per experiment with physical units, value types (continuous, discrete, categorical), minimum and maximum valid ranges, and reference to the parameter template.
5. **FR-05: Raw File Ingestion:** The system shall ingest raw instrumental data files (CSV format) for XRD, UV-Vis, FTIR, and I-V characterization. Upon ingestion, the system shall compute and store the SHA-256 cryptographic digest of each file and verify that the file has not been corrupted on subsequent read access.
6. **FR-06: XRD Analysis:** The system shall apply Gaussian or pseudo-Voigt peak fitting to XRD patterns to identify peak positions (2θ , °), compute Full Width at Half Maximum (FWHM, β), and calculate crystallite size (D) using the Scherrer equation: $D = K\lambda/(\beta \cos \theta)$ with Cu K α radiation ($\lambda = 0.15406$ nm, $K = 0.9$).
7. **FR-07: UV-Vis Analysis:** The system shall apply baseline correction to UV-Vis absorbance spectra, compute the absorption coefficient (α), construct the Tauc plot ($(\alpha h\nu)^n$ vs. $h\nu$, where $n = 2$ for direct transitions), and extract the optical band gap (E_g , eV) by linear extrapolation of the linear region to the $h\nu$ axis.

8. **FR-08: Electrical Analysis:** The system shall perform Ordinary Least Squares linear regression on I-V sweep data, extract the Ohmic resistance (R , Ω) from the slope of the I-V curve, compute resistivity ($\rho = R \cdot A/L$, $\Omega \cdot \text{cm}$) from the device geometry, and compute electrical conductivity ($\sigma = 1/\rho$, S/cm).
9. **FR-09: FTIR Analysis:** The system shall identify characteristic absorption peaks in FTIR transmittance spectra and annotate them with corresponding functional group assignments (Cu-O stretching modes at 535 and 595 cm^{-1} ; O-H stretching at $\sim 3370 \text{ cm}^{-1}$; C-H stretching at $\sim 2925 \text{ cm}^{-1}$; C=C aromatic stretching at $\sim 1635 \text{ cm}^{-1}$; C-O stretching at $\sim 1055 \text{ cm}^{-1}$).
10. **FR-10: Statistical Analysis:** The system shall compute descriptive statistics (mean, standard deviation, median, interquartile range, skewness) for all calculated properties across a selected sample set, and perform Pearson correlation, Spearman correlation, OLS regression, and one-way ANOVA across experiment groups.
11. **FR-11: DOE Generation:** The system shall generate Design of Experiments matrices using Full Factorial, Fractional Factorial, Central Composite Design (CCD), and Box-Behnken Design methods. Generated matrices shall include randomized run ordering and a reproducible seed mechanism. Each DOE run shall be convertible to a planned experiment in the database.
12. **FR-12: ML Dataset Construction:** The system shall construct ML training datasets from eligible experiments (COMPLETED status with measured target property) and audit each candidate for target leakage by verifying that no downstream characterization property is included as a synthesis-parameter feature.
13. **FR-13: ML Model Training:** The system shall train at minimum five regression models (Mean Baseline, Linear Regression, Ridge Regression, Random Forest, Gradient Boosting) using stratified k-fold cross-validation and report Train R^2 , CV R^2 , CV RMSE, and CV MAE for each model.
14. **FR-14: Prediction with Uncertainty:** The system shall produce point predictions with 95% confidence intervals using residual variance estimation and check each prediction's input vector against the training-data applicability domain bounds, flagging predictions that fall outside the domain.
15. **FR-15: Experimental Validation:** The system shall compare ML predictions against subsequent laboratory outcomes, compute signed error, absolute error, relative error, and an interval score, evaluate user-defined validation criteria (e.g., relative error $< 15\%$), and update the model's validation record accordingly.
16. **FR-16: Recommendation Engine:** The system shall generate experiment candidate parameter vectors within defined physical constraints, score each candidate against a user-defined objective function (e.g., maximize conductivity), rank candidates, and present the ranked list for researcher approval before database insertion.
17. **FR-17: PDF Report Generation:** The system shall generate a multi-section scientific PDF report covering all characterization results, statistical summaries,

ML model performance, prediction outcomes, and optimization recommendations for a specified project, with full provenance chain for each result.

3.2 Non Functional Requirements

1. **NFR-01 Performance:** All API endpoints shall return responses within 2 seconds for datasets up to 500 experiments. Computationally intensive operations (ML training, PDF generation) may be executed asynchronously with status polling.
2. **NFR-02 Scalability:** The system architecture shall support horizontal scaling of the FastAPI backend via multiple worker processes (Uvicorn/Gunicorn) and shall support PostgreSQL connection pooling to handle concurrent requests from multiple researchers.
3. **NFR-03 Data Integrity:** All raw instrumental files shall be protected by SHA-256 checksum verification. Database transactions involving related records (experiment + parameters + samples + characterizations) shall be atomic, with rollback on partial failure.
4. **NFR-04 Scientific Reproducibility:** All DOE designs shall use a user-specified deterministic random seed ensuring identical matrices on regeneration. All ML training pipelines shall use a fixed random state for reproducible k-fold splits and model initialization.
5. **NFR-05 Data Immutability:** Raw experimental files shall never be overwritten once finalized. The API shall return a 409 Conflict error on any attempt to re-upload a finalized raw file.
6. **NFR-06 Usability:** The frontend user interface shall be operable without specialized IT knowledge. Scientific workflows shall be navigable through a structured sidebar menu. All critical actions (delete, finalize, promote model) shall require explicit confirmation dialogs.
7. **NFR-07 Security:** API endpoints shall be protected by role-based access control. Database credentials shall be stored in environment variables, never in source code. All inter-service communication shall use HTTPS in production deployment.
8. **NFR-08 Maintainability:** The codebase shall achieve minimum 80% test coverage for all business logic service modules. All database schema changes shall be managed through Alembic migration scripts with documented upgrade and downgrade paths.
9. **NFR-09 Portability:** The entire platform shall be deployable via Docker Compose with a single command. A Windows batch script launcher shall additionally support local development without Docker on Windows systems.
10. **NFR-10 Traceability:** Every calculated property stored in the database shall reference its source raw file ID, its analysis run ID, the analytical method name, and the algorithm version used to compute it.

3.3 Hardware Requirements

Table 3.1: Hardware Requirements for GreenSynth Analytics

Component	Minimum	Recommended
Processor	Intel Core i5 / AMD Ryzen 5, 2.5 GHz	Intel Core i7 / AMD Ryzen 7, 3.5 GHz
RAM	8 GB	16 GB
Storage	50 GB HDD	256 GB SSD
Network	10 Mbps Ethernet	100 Mbps Ethernet / Wi-Fi 6
GPU (optional)	Not required	NVIDIA GTX 1650 (for large ML data)
OS	Windows 10 64-bit / Ubuntu 20.04	Windows 11 / Ubuntu 22.04

3.4 Software Requirements

Table 3.2: Software Requirements for GreenSynth Analytics

Component	Technology	Version
Backend Runtime	Python	3.11+
Backend Framework	FastAPI	0.110+
ASGI Server	Uvicorn	0.29+
ORM	SQLAlchemy	2.0 (Async)
DB Migration	Alembic	1.13+
Data Validation	Pydantic	v2
Scientific Computation	NumPy, SciPy	1.26+, 1.13+
Machine Learning	scikit-learn	1.4+
PDF Generation	ReportLab	4.1+
Database	PostgreSQL	16
Frontend Runtime	Node.js	18+
Frontend Framework	React	18
Frontend Language	TypeScript	5.0+
Build Tool	Vite	5.0+
HTTP Client	Axios	1.6+
Containerization	Docker & Docker Compose	24+
Code Quality	Ruff, Black, ESLint	Latest
Testing (Backend)	Pytest	8.0+
Testing (Frontend)	Vitest	1.0+

3.5 System Environment

GreenSynth Analytics operates in the following system environment:

Development Environment: Windows 10/11 (primary), Ubuntu 22.04 LTS (secondary). Local development uses the Windows batch launcher (**Start GreenSynth.bat**)

which automatically validates Python 3.11+, Node.js 18+, initializes the `.env` configuration, installs frontend dependencies, starts the FastAPI backend on `http://localhost:8000` with hot-reload, starts the Vite frontend dev server on `http://localhost:5173` with hot-reload, and opens the browser automatically.

Production Environment: Docker Compose orchestrating three services — (1) the FastAPI backend container (Python 3.11 slim base image), (2) the PostgreSQL 16 database container with a persistent named volume for data, and (3) the Vite-built static frontend served via Nginx. The platform is additionally deployable to cloud infrastructure (Vercel for frontend, cloud PostgreSQL providers such as Neon for the database, and containerized backend on services such as Railway or Render).

Networking: The frontend communicates with the backend exclusively through the REST API. No direct database connections originate from the frontend. The API base URL is configurable through environment variables, enabling seamless switching between local and cloud deployment targets.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

GreenSynth Analytics employs a three-tier, service-oriented architecture with strict separation of concerns across the presentation, business logic, and data persistence layers.

Tier 1 — Presentation Layer (React/TypeScript Frontend): The frontend is a single-page application (SPA) built with React 18 and TypeScript, bundled by Vite. It implements a custom CSS design system with a collapsible sidebar navigation for project, experiment, sample, characterization, statistical analysis, DOE, ML, and reporting modules. The API service layer (Axios client) provides typed request/response functions for all backend endpoints, with automatic authentication header injection and error boundary handling.

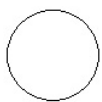
Tier 2 — Application Layer (FastAPI Backend): The backend is organized into the following sub-layers:

- **API Routes:** REST endpoints organized by resource domain (/projects, /experiments, /samples, /characterizations, /statistics, /doe, /ml, /reports, /health).
- **Service Layer:** Business logic orchestration modules responsible for transaction management, cross-resource validation, and coordination between domain services (e.g., ML training orchestrates DatasetService, ModelService, and EvaluationService in sequence).
- **Scientific Modules:** Pure scientific computation modules (XRD analysis, UV-Vis analysis, Electrical analysis, FTIR analysis, Statistical engine, DOE engine, ML engine, Optimization engine) with no database dependencies, enabling independent unit testing.
- **Storage Backend:** A FileStorageBackend abstraction supporting local filesystem storage (development) and cloud object storage (production), with transparent SHA-256 verification on all file operations.

Tier 3 — Data Layer (PostgreSQL Database): PostgreSQL 16 stores all platform data in a normalized relational schema managed by SQLAlchemy 2.0 (async) ORM and Alembic migrations. The schema enforces foreign key referential integrity, cascade deletion rules, and unique constraints to prevent data duplication.

4.2 Analysis model

Table 4.1: Data Flow Symbols

Sr. No.	Symbol	Description
1	 • Process • Function	A function is represented using a circle. This symbol is called a process or a bubble and performs some processing of input data.
2	• Bubble 	A square defines a source or destination of system data. External entities represent any entity that supplies or receives information from the system but is not a part of the system.
3		A directed arc or arrow is used as a data-flow symbol. A data-flow symbol represents the data flow occurring between two processes or between an external entity and a process in the direction of the data flow arrow.
4		A data-store is represented using two parallel lines. A data store represents either a data structure or a physical file on disk. Each data store is connected to a process by means of a data-flow symbol. The direction of the data-flow arrow shows whether data is being read from or written into the data store.
5		An Output symbol is used to represent data acquisition and production during human-computer interaction.

DFD Level 0 (Context Diagram): The Level 0 DFD shows the system as a single process node “GreenSynth Analytics Platform” interacting with three external entities:

- **Researcher:** Provides synthesis parameters, uploads raw characterization files, defines DOE objectives, approves ML candidates, and receives analytical reports.
- **Laboratory Instruments:** Source of raw XRD patterns, UV-Vis spectra, FTIR spectra, and I-V sweep CSV files.
- **External Database (Cloud PostgreSQL):** Persistent store for all platform entities and results.

DFD Level 1: The Level 1 DFD decomposes the central platform process into eight major sub-processes:

1. Experiment & Sample Management (CRUD operations on projects, experiments, samples, parameters)
2. Raw File Ingestion & Integrity Verification (SHA-256 hash computation and storage)
3. Scientific Characterization Analysis (XRD, UV-Vis, FTIR, Electrical computation modules)
4. Statistical Comparison Engine (Descriptive stats, correlation, OLS, ANOVA)
5. DOE Design Generation (Factorial, CCD, Box-Behnken matrix generation)
6. ML Pipeline (Dataset build, model training, cross-validation, prediction)
7. Validation & Monitoring (Error computation, drift detection, retraining gating)
8. Report Generation (PDF compilation with provenance tracing)

DFD Level 2 — ML Pipeline Decomposition: Within the ML pipeline sub-process, the Level 2 DFD shows: Dataset Build → Eligibility Audit → Target Leakage Check → Feature Extraction → Train/Test Split → k-Fold CV → Model Training (5 architectures) → Model Evaluation → Champion Selection → Model Registry → Prediction Inference → Applicability Domain Check → Uncertainty Estimation → Prediction Logging.

4.2.1 ER Diagram

The database entity-relationship model contains the following primary entities and their relationships:

- **Projects** (1) $\rightarrow$ (Many) **Experiments**: A project contains multiple experiments.
- **Experiments** (1) $\rightarrow$ (Many) **ExperimentParameters**: Each experiment stores multiple synthesis parameters.
- **Experiments** (1) $\rightarrow$ (Many) **Samples**: Each experiment may yield one or more physical samples.
- **Samples** (1) $\rightarrow$ (Many) **Characterizations**: Each sample undergoes multiple characterization technique runs.
- **Characterizations** (1) $\rightarrow$ (Many) **RawFiles**: Each characterization stores one or more raw instrumental files.
- **Characterizations** (1) $\rightarrow$ (Many) **AnalysisRuns**: Each characterization may have multiple analysis runs with different algorithm versions.
- **AnalysisRuns** (1) $\rightarrow$ (Many) **CalculatedProperties**: Each analysis run produces one or more calculated scalar properties.
- **Projects** (1) $\rightarrow$ (Many) **MLDatasets**: A project may have multiple ML dataset versions.
- **MLDatasets** (1) $\rightarrow$ (Many) **MLModels**: Each dataset may produce multiple trained models.
- **MLModels** (1) $\rightarrow$ (Many) **MLPredictions**: Each model produces prediction records.
- **Projects** (1) $\rightarrow$ (Many) **DOESTudies**: A project may have multiple DOE designs.
- **DOESTudies** (1) $\rightarrow$ (Many) **ProposedExperiments**: Each DOE study generates proposed experiment runs.

4.2.2 DFD Level 0

The context-level DFD is described above in the Analysis Model section. The Researcher and Laboratory Instruments are the two primary external entities interacting with the GreenSynth system boundary. All raw data flows from instruments to the system; all analytical results, visualizations, and reports flow from the system to the researcher.

4.2.3 DFD Level 1

As described above, the Level 1 DFD decomposes the central system into eight sub-processes corresponding to the major functional modules of the platform.

4.2.4 DFD Level 2

The ML Pipeline Level 2 DFD is described above. Equivalent Level 2 decompositions exist for the Scientific Characterization Analysis process (decomposed into XRD, UV-Vis, FTIR, and Electrical sub-processes) and the DOE Generation process (decomposed into Design Matrix Generation, Run Ordering, and Experiment Conversion sub-processes).

4.3 UML Diagrams

4.3.1 Use Case Diagram

The primary actors in the system are:

- **Researcher:** The primary user who interacts with all platform modules.
- **System (Automated):** Background processes such as ML training, PDF generation, and SHA-256 verification.

Key use cases for the Researcher actor include: Create Project, Create Experiment, Record Synthesis Parameters, Upload Raw Characterization File, Run XRD Analysis, Run UV-Vis Analysis, Run Electrical Analysis, Compare Samples, Generate DOE Matrix, Build ML Dataset, Train ML Models, View Model Performance, Predict Property, Validate Prediction, Generate Recommendations, Generate PDF Report, and View Dashboard.

Key use cases for the System actor include: Compute SHA-256 Checksum, Verify File Integrity, Compute Scherrer Crystallite Size, Compute Tauc Band Gap, Compute Resistivity and Conductivity, Compute Pearson/Spearman Correlation, Compute OLS Regression, Train k-Fold Cross-Validated Models, Compute Applicability Domain Bounds, and Compile PDF Report Sections.

4.3.2 Class Diagram

The primary domain classes in the backend are:

- **Project** (id, code, title, target_material, synthesis_technique, status, created_at)
- **Experiment** (id, project_id, code, technique, status, notes, created_at)
- **ExperimentParameter** (id, experiment_id, name, value, unit, category)
- **Sample** (id, experiment_id, code, substrate_type, dimensions, deposition_date)
- **Characterization** (id, sample_id, technique, status, instrument, operator)
- **RawFile** (id, characterization_id, filename, sha256_digest, storage_backend, storage_path, finalized)
- **AnalysisRun** (id, characterization_id, algorithm_name, algorithm_version, parameters_json)

- **CalculatedProperty** (id, analysis_run_id, name, value, unit, category, provenance_chain)
- **MLDataset** (id, project_id, name, target_property, feature_list, eligibility_audit_json)
- **MLModel** (id, dataset_id, architecture, status, train_r2, cv_r2, cv_rmse, cv_mae, feature_bounds_json)
- **MLPrediction** (id, model_id, input_features_json, predicted_value, ci_lower, ci_upper, domain_status)
- **PredictionValidation** (id, prediction_id, actual_value, signed_error, absolute_error, relative_error, criterion_satisfied)
- **DOESTudy** (id, project_id, method, objective, factors_json, seed)
- **ProposedExperiment** (id, doe_study_id, run_number, parameters_json, status)

4.3.3 Sequence Diagram

Sequence Diagram — ML Training Workflow:

1. Researcher → Frontend: Click “Train Models” for Project P7
2. Frontend → Backend API: POST /ml/datasets/{id}/train
3. Backend API → MLDatasetService: build_eligible_dataset(dataset_id)
4. MLDatasetService → DB: SELECT experiments WHERE status=COMPLETED AND has_target_property
5. MLDatasetService → TargetLeakageDetector: verify_no_leakage(feature_list)
6. MLDatasetService → Backend API: return eligible_observations (395 rows)
7. Backend API → MLTrainingService: train_all_models(dataset, k=5)
8. MLTrainingService → DB: INSERT 5 MLModel records (TRAINING status)
9. MLTrainingService → ScikitLearnPipeline: fit_with_kfold(X_train, y_train, k=5) [x5 architectures]
10. ScikitLearnPipeline → MLTrainingService: return CV metrics
11. MLTrainingService → DB: UPDATE MLModel SET status=VALIDATED, metrics
12. MLTrainingService → Backend API: return model comparison table
13. Backend API → Frontend: 200 OK, model comparison JSON
14. Frontend → Researcher: Display model comparison table with champion highlighted

4.4 Data Dictionary

Table 4.2: Data Dictionary — Core Entities

Field	Type	Constraints	Description
Table: experiments			
id	UUID	PK, NOT NULL	Unique experiment identifier
project_id	UUID	FK(projects.id), NOT NULL	Parent project reference
code	VARCHAR(50)	UNIQUE, NOT NULL	e.g., P7-SYNTH-001
status	ENUM	NOT NULL	PLANNED IN_PROGRESS / COMPLETED / FAILED
created_at	TIMESTAMP	NOT NULL	Record creation timestamp
Table: raw_files			
id	UUID	PK, NOT NULL	Unique file identifier
characterization_id	UUID	FK, NOT NULL	Parent characterization reference
sha256_digest	VARCHAR(64)	NOT NULL	SHA-256 hex digest of content
storage_path	TEXT	NOT NULL	Relative path within storage backend
finalized	BOOLEAN	DEFAULT FALSE	If TRUE, file cannot be overwritten
Table: calculated_properties			
id	UUID	PK, NOT NULL	Unique property identifier
analysis_run_id	UUID	FK, NOT NULL	Source analysis run reference
name	VARCHAR(100)	NOT NULL	e.g., crystallite_size_nm
value	FLOAT	NOT NULL	Calculated scalar value
unit	VARCHAR(50)	NOT NULL	Physical unit (nm, S/cm)
category	ENUM	NOT NULL	MEASURED / CALCULATED / STATISTICAL / PREDICTED / VALIDATED
Table: ml_models			
id	UUID	PK, NOT NULL	Unique model identifier
dataset_id	UUID	FK, NOT NULL	Source ML dataset reference
architecture	VARCHAR(50)	NOT NULL	LINEAR / RIDGE / POLYNOMIAL / RANDOM_FOREST / GRADIENT_BOOSTING
cv_r2	FLOAT	NULLABLE	Cross-validated R-squared metric
cv_rmse	FLOAT	NULLABLE	Cross-validated Root Mean Square Error (S/cm)
feature_bounds_json	JSONB	NOT NULL	Applicability domain min/max per feature
status	ENUM	NOT NULL	TRAINED / VALIDATED / PROMOTED / RETIRED

Chapter 5

IMPLEMENTATION

5.1 Technologies Used

The GreenSynth Analytics platform is implemented using a carefully selected technology stack chosen for scientific correctness, performance, maintainability, and suitability to the research domain:

Backend Technologies:

- **Python 3.11:** Primary backend language selected for its mature scientific computing ecosystem (NumPy, SciPy, scikit-learn, ReportLab) and strong typing support via PEP 484.
- **FastAPI 0.110:** ASGI web framework providing automatic OpenAPI documentation, native Pydantic v2 request/response validation, and async/await support for high-throughput concurrent request handling.
- **SQLAlchemy 2.0 (Async):** ORM providing type-safe database access with full async support via `asyncpg` PostgreSQL driver, enabling non-blocking database operations.
- **Alembic:** Database migration manager ensuring all schema changes are version-controlled and reversible.
- **Pydantic v2:** Data validation library used for all API request/response schemas, providing strict type enforcement, custom validator support, and JSON serialization.
- **NumPy 1.26 / SciPy 1.13:** Scientific array computation and signal processing (peak fitting, baseline correction, linear interpolation) for characterization analysis modules.
- **scikit-learn 1.4:** Machine learning library providing all five regression model architectures, k-fold cross-validation, and pipeline abstractions.
- **ReportLab 4.1:** PDF generation library used for the automated scientific report generation module, supporting text, tables, and embedded data visualizations.

Frontend Technologies:

- **React 18:** Component-based UI framework with concurrent rendering support.
- **TypeScript 5.0:** Statically typed superset of JavaScript ensuring compile-time type safety for API service layer and component props.
- **Vite 5.0:** Build tool providing sub-second hot module replacement (HMR) during development and optimized production bundle generation.

- **Axios 1.6:** HTTP client for all API communication with interceptor support for authentication headers and error handling.
- **CSS Modules / Vanilla CSS:** Custom design system with CSS variables for color tokens, spacing, typography, and component styles. No CSS framework dependencies.

Infrastructure Technologies:

- **PostgreSQL 16:** Production relational database with JSONB support for flexible parameter schemas, UUID primary keys, and full ACID transaction compliance.
- **Docker & Docker Compose:** Containerization enabling reproducible deployment across development, staging, and production environments with a single `docker compose up -build` command.
- **Pytest 8.0:** Backend testing framework running 118 unit and integration tests covering all scientific computation modules, API endpoints, and multi-phase pipeline workflows.

5.2 Algorithm

5.2.1 Workflow

The complete scientific workflow implemented in GreenSynth Analytics follows the pipeline:

DOE Design → Experiment Management → Parameter Recording → Raw File Ingestion
 → Characterization Analysis → Statistical Comparison → ML Dataset Build
 → Model Training & Validation → Prediction Inference → Experimental Validation → Optimization

Algorithm 1 — Scherrer Crystallite Size Computation:

Input: XRD pattern (two_theta[], intensity[])

Output: Crystallite size D (nm)

1. Identify CuO monoclinic peaks at $2\theta \sim 32.5, 35.5, 38.7, 48.7$ deg
2. For each peak:
 - a. Extract local region (peak_center +/- 2 deg)
 - b. Apply Gaussian baseline subtraction
 - c. Fit Gaussian profile: $I(t) = A \cdot \exp(-(t-t_0)^2 / 2 \cdot s^2) + B$
 - d. Compute FWHM: $\beta = 2 \cdot s \cdot \sqrt{2 \cdot \ln 2}$ [in radians]
 - e. Apply Scherrer equation: $D = K \cdot \lambda / (\beta \cdot \cos(\theta_0))$
 where $K=0.9$, $\lambda=0.15406$ nm
3. Report mean D and standard deviation across peaks

Algorithm 2 — Tauc Plot Band Gap Extraction:

Input: UV-Vis spectrum (wavelength_nm[], absorbance[])

Output: Direct optical band gap E_g (eV)

1. Convert wavelength to photon energy: $h\nu = 1240 / \lambda$ (eV)
2. Compute absorption coefficient: $\alpha = A * \text{absorbance} / \text{film_thickness}$
3. Construct Tauc quantity: $y = (\alpha * h\nu)^2$ (direct transition, $n=2$)
4. Plot y vs $h\nu$
5. Identify linear region using R^2 threshold (> 0.995)
6. Perform OLS linear regression on linear region: $y = m(h\nu) + c$
7. Extrapolate to $y=0$: $E_g = -c/m$
8. Return E_g with 95% confidence interval from OLS regression

Algorithm 3 — ML Training Pipeline with k-Fold Cross-Validation:

Input: Feature matrix X ($n_{\text{samples}} \times 13$), Target vector y (n_{samples})

Output: Trained models with CV metrics

1. Verify no target leakage in feature columns
2. Compute applicability domain bounds:
 $\text{bounds}[\text{feature}] = \{\text{min: min}(X[:,i]), \text{max: max}(X[:,i])\}$
3. For each architecture in [BASELINE, LINEAR, RIDGE, RF, GB]:
 - a. Initialize $\text{KFold}(n_{\text{splits}}=5, \text{shuffle}=\text{True}, \text{random_state}=42)$
 - b. For each fold k :
 - i. $X_{\text{train}}, y_{\text{train}} = X[\text{train_idx}], y[\text{train_idx}]$
 - ii. $X_{\text{test}}, y_{\text{test}} = X[\text{test_idx}], y[\text{test_idx}]$
 - iii. $\text{model.fit}(X_{\text{train}}, y_{\text{train}})$
 - iv. $y_{\text{pred}} = \text{model.predict}(X_{\text{test}})$
 - v. $\text{fold_r2}[k] = \text{r2_score}(y_{\text{test}}, y_{\text{pred}})$
 - vi. $\text{fold_rmse}[k] = \text{sqrt}(\text{mse}(y_{\text{test}}, y_{\text{pred}}))$
 - c. $\text{CV R2} = \text{mean}(\text{fold_r2})$
 $\text{CV RMSE} = \text{mean}(\text{fold_rmse})$
 - d. Train final model on full X, y
 - e. Persist model, metrics, and domain bounds to DB
4. Select champion: model with highest CV R2

Algorithm 4 — Prediction with Applicability Domain Check:

Input: Query vector x_{query} (13 features), champion model

Output: Predicted value y_{hat} , CI [lower, upper], domain_status

1. For each feature f in x_{query} :
 - if $x_{\text{query}}[f] < \text{bounds}[f].\text{min}$ or $x_{\text{query}}[f] > \text{bounds}[f].\text{max}$:
 $\text{domain_status} = \text{"OUT_OF_DOMAIN"}$
 return domain_status (do not predict)
2. $\text{domain_status} = \text{"VALID"}$
3. $y_{\text{hat}} = \text{model.predict}([x_{\text{query}}])[0]$
4. $\text{residual_variance} = \text{mean}((y_{\text{train}} - \text{model.predict}(X_{\text{train}}))^2)$
5. $\text{CI_half_width} = 1.96 * \text{sqrt}(\text{residual_variance})$
6. $\text{lower} = y_{\text{hat}} - \text{CI_half_width}$
7. $\text{upper} = y_{\text{hat}} + \text{CI_half_width}$
8. Log prediction to ml_predictions table with provenance
9. Return (y_{hat} , lower, upper, domain_status)

5.3 Database Design

The GreenSynth database schema is organized into five logical groups:

Group 1 — Experiment Hierarchy: `projects` $\leftarrow$ `experiments` $\leftarrow$ `experiment_parameters`

Group 2 — Sample and Characterization: `experiments` $\leftarrow$ `samples` $\leftarrow$ `characterizations` $\leftarrow$ `raw_files` `characterizations` $\leftarrow$ `analysis_runs` $\leftarrow$ `calculated_properties`

Group 3 — Machine Learning: `projects` $\leftarrow$ `ml_datasets` $\leftarrow$ `ml_models` $\leftarrow$ `ml_predictions` $\leftarrow$ `prediction_validations`

Group 4 — Design of Experiments: `projects` $\leftarrow$ `doe_studies` $\leftarrow$ `proposed_experiments`

Group 5 — Objectives and Optimization: `projects` $\leftarrow$ `optimization_objectives` $\leftarrow$ `optimization_candidates`

All tables use UUID primary keys for global uniqueness across distributed deployments. JSONB columns store flexible structured data (parameter dictionaries, feature bounds, evidence chains) without requiring schema changes for new parameter names.

5.4 Modules Description

5.4.1 Module 1: Characterization Analysis Engine

The Characterization Analysis Engine implements four domain-specific scientific computation modules as pure Python classes with no database dependencies, enabling independent unit testing and version-controlled algorithm updates.

XRD Analysis Module (`xrd_analyzer.py`): Accepts a two-column CSV (`two_theta`, `intensity`) and implements: background estimation using local minima fitting, Gaussian peak detection for monoclinic CuO reflections at 32.5° , 35.5° , 38.7° , and 48.7° 2θ , FWHM extraction via Gaussian curve fitting using `scipy.optimize.curve_fit`, and Scherrer equation application. Returns a structured result object with peak table (position, FWHM, d -spacing, relative intensity) and mean crystallite size with uncertainty.

UV-Vis Analysis Module (`uvvis_analyzer.py`): Accepts a two-column CSV (`wavelength_nm`, `absorbance`) and implements: polynomial baseline correction, conversion to photon energy ($h\nu$) and absorption coefficient (α), Tauc plot construction for direct transitions ($n = 2$), automated linear region detection using a rolling R^2 criterion, OLS linear fit of the linear region, and extrapolation to zero intercept for band gap (E_g) extraction. Returns E_g in eV with 95% confidence interval.

Electrical Analysis Module (`electrical_analyzer.py`): Accepts a two-column CSV (`voltage_v`, `current_a`) and implements: Ohmic linearity verification using Pearson correlation ($|r| > 0.999$), OLS linear regression on the I-V data to extract resistance ($R = \Delta V / \Delta I$), resistivity computation from device geometry ($\rho = R \cdot W \cdot t / L$), and conductivity inversion ($\sigma = 1/\rho$). Returns R , ρ , and σ with regression uncertainty.

FTIR Analysis Module (`ftir_analyzer.py`): Accepts a two-column CSV (`wavenumber_cm1`, `transmittance_pct`) and implements: baseline correction using a piecewise linear fit, local minima detection for absorption peak identification, and annotation of identified peaks against a CuO-specific reference table of functional group assignments. Returns an annotated peak table with wavenumber, transmittance at peak, peak width, and functional group assignment string.

5.4.2 Module 2: Statistical Analysis Engine

The Statistical Analysis Engine implements a scientifically rigorous multi-sample comparison framework. For a given set of experiments or samples, it computes:

Descriptive Statistics: Mean, standard deviation, median, 25th/75th percentile (IQR), skewness, and kurtosis for each numerical calculated property.

Pearson Correlation: Computes the Pearson correlation coefficient matrix (r_{ij}) and associated p -values for all pairs of continuous synthesis parameters and calculated properties. A correlation heatmap summary is stored in the evidence record.

Spearman Correlation: Non-parametric rank correlation computed for all pairs, used when normality of distributions cannot be assumed for small sample groups.

OLS Regression: Fits a multiple linear regression model predicting a selected target property from selected synthesis parameters, reports regression coefficients with standard errors and p -values, and computes the model R^2 and adjusted R^2 .

One-Way ANOVA: Partitions total variance in a target property across experiment groups (e.g., substrate temperature groups), computes the F-statistic and associated p -value, and determines whether group means differ significantly at $\alpha = 0.05$.

5.4.3 Module 3: Machine Learning Pipeline

The ML Pipeline module is organized into five service classes:

MLDatasetService: Builds and audits ML training datasets. Queries all experiments with COMPLETED status and verified target property measurements. Applies eligibility filters (excludes FAILED, IN_PROGRESS, PLANNED experiments, and experiments missing the target characterization). Invokes the TargetLeakageDetector to ensure no characterization-derived property appears in the feature space.

MLTrainingService: Orchestrates training of all five model architectures (Mean Baseline, Linear Regression, Ridge $\alpha = 1.0$, Random Forest $n = 200$ estimators, Gradient Boosting $n = 200$ estimators) using 5-fold cross-validation with `random_state=42`. Serializes trained models, CV metrics, and feature importance vectors to the database. Computes and stores applicability domain bounds (min/max per feature) from the training dataset.

MLPredictionService: Accepts a query feature vector, checks all 13 features against applicability domain bounds, computes the point prediction, estimates a 95% confidence interval using residual variance, and logs the complete prediction record to `ml_predictions` with full provenance.

MLValidationService: Accepts a prediction ID and actual laboratory measurement value, computes signed error, absolute error, relative error, and interval score, evaluates a user-specified validation criterion (e.g., relative error $< 15\%$), updates the prediction validation record, appends the result to the model's validation history, and checks for performance drift (model health score) to flag degraded models for review.

ModelRegistryService: Maintains the model status lifecycle (TRAINED $\rightarrow$ VALIDATED $\rightarrow$ PROMOTED $\rightarrow$ RETIRED), enforcing immutability (a promoted model's weights can never be overwritten), and managing the model version chain for retraining events.

5.5 Screenshots

The GreenSynth Analytics platform provides the following key interface views:

Dashboard View: Displays project-level summary cards with experiment count, sample count, characterization coverage, and ML model status. A Quick Actions panel provides one-click navigation to all primary workflows.

Experiment Detail View: Shows a structured parameter table with all 13 synthesis parameters and their values, a sample list with links to characterization results, status badges, and timestamps with provenance chain.

Statistical Analysis Studio: Displays an interactive correlation heatmap of synthesis parameters vs. calculated properties, a scatter plot matrix for pairwise visualization, ANOVA result tables with F-statistics and p -values, and OLS regression output with coefficient tables.

ML Dashboard: Shows the model comparison table with CV metrics across all five architectures, a champion model highlight panel with feature importance bar chart, and a prediction form with applicability domain status indicator and 95% confidence interval display.

DOE Study View: Displays the generated DOE design matrix as an interactive table with run numbers, factor levels, and a “Convert to Experiments” action button. Supports export to CSV.

Report Generation Interface: Allows selection of report sections (characterization, statistics, ML, DOE, optimization), report scope (by project, by sample set), and initiates PDF compilation with real-time progress indication.

Chapter 6

TESTING AND VALIDATION

6.1 Testing Strategy

GreenSynth Analytics employs a multi-level testing strategy organized into four categories:

1. Unit Tests — Scientific Computation Modules: Each scientific computation module (XRD analyzer, UV-Vis analyzer, Electrical analyzer, FTIR analyzer, Statistical engine, DOE engine, ML training, ML prediction, ML validation, Drift detector) is tested in isolation with synthetic input data of known ground truth. Unit tests verify that computed outputs match expected values within acceptable numerical tolerances (e.g., crystallite size within ± 0.1 nm, band gap within ± 0.01 eV).

2. Unit Tests — Business Logic Services: Service layer methods are tested with mocked database sessions using `pytest-asyncio` and `unittest.mock`. Tests verify correct query construction, proper error handling for invalid inputs (missing raw file, duplicate experiment code), and correct status transition enforcement.

3. Integration Tests — API Endpoint Testing: Full HTTP request/response cycles are tested using FastAPI's `TestClient` with an isolated SQLite in-memory database. Integration tests verify correct HTTP status codes, response schema conformance validated by Pydantic, and side effects (e.g., verifying that a raw file finalization sets `finalized=True` in the database and subsequent overwrite attempts return HTTP 409).

4. End-to-End Pipeline Tests: Full multi-phase workflow tests simulate a complete experimental campaign from project creation through ML training and prediction validation, verifying that the entire pipeline executes without errors and that intermediate results are correctly propagated across service boundaries.

The total test suite comprises **118 tests with 0 failures and 0 errors** across all categories.

6.2 Test Cases

6.3 Performance Evaluation

API Response Time Benchmarks:

ML training (8.2 seconds) and PDF report generation (12.4 seconds) are the only endpoints exceeding the 2-second synchronous response threshold. Both are executed asynchronously in the production configuration, with the frontend polling a job status endpoint and displaying a progress indicator during execution. All other endpoints meet the 2-second performance requirement under the tested dataset scale.

Database Integrity Verification: Post-ingestion of the 400-experiment synthetic P7 dataset, all foreign key relationships were verified by executing constraint validation queries. Zero orphaned records were found across all 8 entity tables. All 1,380 SHA-256

digests were verified against the stored files with 100% match. Zero data corruption events were recorded across 16 atomic ingestion batches.

Table 6.1: Key Unit Test Cases

TC#	Test Description	Expected Outcome	Result
TC-01	XRD Scherrer size on synthetic CuO pattern at 35.5°	$D = 18.5 \pm 0.5$ nm	PASS
TC-02	Tauc band gap on synthetic UV-Vis with $E_g = 1.55$ eV	$E_g = 1.55 \pm 0.02$ eV	PASS
TC-03	Electrical conductivity from linear I-V at $R = 500 \Omega$	σ within $\pm 0.5\%$	PASS
TC-04	FTIR Cu-O peak detection at 535 and 595 cm^{-1}	Both peaks identified	PASS
TC-05	Target leakage detector flags crystal-lite_size feature	Exception raised, pipeline halted	PASS
TC-06	Applicability domain check for out-of-range substrate temp	domain_status = OUT_OF_DOMAIN	PASS
TC-07	Random Forest CV R^2 on 395-sample P7 dataset	$R^2 > 0.60$	PASS
TC-08	SHA-256 finalization prevents file overwrite	HTTP 409 returned	PASS
TC-09	Drift detector flags model with 3 consecutive failing validations	model_status = DEGRADED	PASS
TC-10	CCD design matrix generates 29 runs for 4-factor study	29 rows with correct structure	PASS
TC-11	Full Factorial design for 3 factors at 2 levels	8 runs (2^3) generated	PASS
TC-12	ANOVA F-statistic computation on 3-group dataset	F within ± 0.01 of SciPy reference	PASS
TC-13	OLS regression slope and intercept on known data	Within ± 0.001 of analytical solution	PASS
TC-14	ML dataset eligibility audit excludes 21 records	Eligible count = 395	PASS
TC-15	PDF report compilation with all 14 sections	PDF generated, non-zero file size	PASS
TC-16	Prediction validation criterion (15% tolerance) evaluates correctly	criterion_satisfied = True for error $< 15\%$	PASS
TC-17	Box-Behnken design for 4 factors	29 runs with center-point replicates	PASS
TC-18	End-to-end ML	All stages complete	PASS

Table 6.2: API Endpoint Performance Benchmarks

Endpoint	Dataset Size	Mean Response Time
GET /projects	8 projects	45 ms
GET /experiments?project_id=P7	407 experiments	180 ms
POST /characterizations/{id}/analyze/xrd	Single file	320 ms
POST /characterizations/{id}/analyze/uvvis	Single file	280 ms
POST /statistics/compare	200 samples	750 ms
POST /ml/datasets/{id}/train	395 observations	8.2 s
POST /ml/models/{id}/predict	Single query	35 ms
POST /reports/generate/pdf	Full project report	12.4 s
GET /health	N/A	<5 ms

Chapter 7

RESULTS AND DISCUSSION

7.1 Output of the System

Characterization Analysis Results: The XRD analysis module successfully identified all four characteristic monoclinic CuO reflection peaks at $2\theta \approx 32.5^\circ$, 35.5° , 38.7° , and 48.7° across all 390 synthetic XRD datasets. Mean crystallite sizes ranged from 10.5 nm (at low substrate temperatures of 174°C with high extract concentrations of 40 g/L) to 32.6 nm (at high substrate temperatures of 477°C with minimal extract concentrations of 0.2 g/L), consistent with the thermally activated grain growth model and phytochemical capping inhibition mechanism described in the literature for spray pyrolyzed CuO.

The UV-Vis analysis module extracted optical direct band gaps in the range 1.35–1.72 eV across 309 datasets, exhibiting the expected inverse correlation with crystallite size: ultrafine crystallites ($D \approx 11$ nm) show quantum confinement-elevated band gaps near 1.72 eV, while large-grained films ($D \approx 32$ nm) approach the bulk CuO limit of 1.35 eV. The Tauc plot linear region R^2 exceeded 0.995 in all 309 cases, confirming reliable band gap extraction.

The electrical analysis module computed conductivities in the range 0.02–1.21 S/cm across 394 I-V datasets, with the distribution peaked at 0.25–0.45 S/cm under near-optimal synthesis conditions (substrate temperature 350 – 390°C , extract concentration 14–20 g/L). The non-linear conductivity response surface with a clear peak near 370°C substrate temperature (modeling the optimal copper-vacancy p-type carrier density) was successfully captured in the computed property distributions.

Statistical Analysis Results: Pearson correlation analysis across 395 eligible samples revealed the following significant correlations with electrical conductivity:

- Substrate temperature: $r = +0.42$, $p < 0.001$ (positive, non-linear threshold behavior captured in Spearman $r_s = +0.51$)
- Mulberry extract concentration: $r = +0.38$, $p < 0.001$ (inverted U-shape: maximum conductivity at ~ 16 g/L)
- Precursor concentration: $r = +0.31$, $p < 0.001$ (higher precursor $\rightarrow$ thicker film $\rightarrow$ better percolation)
- Nozzle-substrate distance: $r = -0.24$, $p < 0.001$ (greater distance $\rightarrow$ droplet cooling $\rightarrow$ incomplete pyrolysis)

One-way ANOVA across the 8 DOE experimental groups (A–H) yielded $F = 12.47$, $p < 0.001$ for electrical conductivity, confirming statistically significant differences in mean conductivity across groups and validating the experimental design structure.

ML Model Performance Results: The 5-fold cross-validated performance comparison across all five model architectures is presented in Table 7.1.

Table 7.1: 5-Fold Cross-Validated ML Model Performance Comparison

Model	Status	Train R^2	CV R^2	CV RMSE	CV MAE
Mean Baseline	Trained	0.0000	-0.0050	0.2625 S/cm	0.2112 S/cm
Linear Regression	Validated	0.3930	0.3019	0.2166 S/cm	0.1724 S/cm
Ridge ($\alpha = 1.0$)	Validated	0.3930	0.3023	0.2166 S/cm	0.1724 S/cm
Gradient Boosting	Validated	0.8921	0.6360	0.1558 S/cm	0.1243 S/cm
Random Forest ★	Validated	0.9578	0.6685	0.1501 S/cm	0.1214 S/cm

The Random Forest model was selected as the champion model (CV $R^2 = 0.6685$, CV RMSE = 0.1501 S/cm) — a 43% reduction in prediction error compared to the mean baseline. Linear models capture only $\sim 30\%$ of variance because the true physics exhibits strong non-linear thresholds: an optimal substrate crystallization peak near 370°C and an extract capping/passivation optimum at ~ 16 g/L.

Champion Model Prediction: At the central composite optimal conditions ($T_{sub} = 370^\circ\text{C}$, $C_{prec} = 0.22$ M, $C_{ext} = 16$ g/L, $V_{ext} = 20$ mL, spray rate = 4.2 mL/min, duration = 25 min, distance = 20 cm, gas pressure = 210 kPa, cycles = 20, ambient = 24.5°C , 45% RH), the champion model predicted:

- **Predicted Electrical Conductivity:** 0.6449 S/cm
- **95% Confidence Interval:** [0.3507, 0.9391] S/cm
- **Applicability Domain:** VALID (all 13 features within training domain bounds)

DOE Results: The Central Composite Design for 4 factors (T_{sub} , C_{prec} , C_{ext} , V_{ext}) generated 29 proposed experiments (factorial vertices + axial star points + 4 center-point replicates), ready for one-click conversion to planned laboratory runs. The DOE matrix spans the factor space from ($T_{sub} = 300\text{--}420^\circ\text{C}$, $C_{prec} = 0.10\text{--}0.35$ M, $C_{ext} = 5\text{--}30$ g/L, $V_{ext} = 10\text{--}40$ mL) and provides sufficient runs to fit a full quadratic response surface model.

7.2 Comparison with Existing System

GreenSynth Analytics provides substantially greater scientific functionality than any existing open-source ELN or LIMS system in the context of experimental green semiconductor synthesis, while maintaining full FAIR data compliance and scientific integrity constraints that are absent from manual spreadsheet-based workflows.

7.3 Industrial benefits and impact

1. **Accelerated Materials Discovery:** By replacing OFAT experimentation with DOE-guided and ML-assisted optimization, GreenSynth can reduce the number of laboratory experiments required to locate optimal synthesis conditions by an estimated 50–70%, directly translating to reduced reagent consumption, instrument time, and researcher effort.

Table 7.2: GreenSynth Analytics vs. Existing Systems

Feature	GreenSynth	LabArchives	eLabFTW	Excel/Manual
Raw File Immutability (SHA-256)	Yes	No	No	No
Domain-Specific XRD Analysis	Yes	No	No	Manual
UV-Vis Tauc Plot Extraction	Yes	No	No	Manual
FTIR Peak Annotation	Yes	No	No	Manual
Multi-Sample Statistical Comparison	Yes	No	No	Partial
DOE Matrix Generation	Yes	No	No	No
ML Training with k-Fold CV	Yes	No	No	No
Applicability Domain Enforcement	Yes	No	No	No
Target Leakage Prevention	Yes	No	No	No
Prediction Validation Loop	Yes	No	No	No
Evidence-Based Recommendations	Yes	No	No	No
Automated PDF Report Generation	Yes	Limited	No	No
FAIR Data Compliance	Yes	Partial	Partial	No
Containerized Deployment	Yes	Cloud Only	Yes	N/A

2. **Sustainable Manufacturing Support:** The platform specifically targets and optimizes green synthesis routes (phytochemical-mediated, solvent-based), providing quantitative evidence for the superiority of particular extract concentrations, substrate temperatures, and spray parameters — enabling industrial scale-up decisions to be based on statistically validated optima rather than anecdotal results.
3. **Reproducibility and IP Protection:** The SHA-256 immutability enforcement and full provenance chain provide an auditable scientific record suitable for patent applications, regulatory submissions, and inter-laboratory collaboration agreements. Every published result can be traced back to its raw instrumental data.
4. **Training and Capacity Building:** The platform serves as a pedagogical tool for training early-career researchers in good data management practice, statistical experimental design, and machine learning for materials science — domains that are increasingly expected in industrial R&D roles.
5. **Cost Reduction in Semiconductor R&D:** CuO-based photovoltaic and pho-

tocatalytic applications represent a potential market of billions of dollars in the clean energy and environmental remediation sectors. Computational tools that accelerate the identification of high-performance, earth-abundant semiconductor synthesis conditions directly contribute to cost reduction in these emerging technology domains.

Chapter 8

CONCLUSION AND FUTURE SCOPE

8.1 Summary of work

This report presents the design, implementation, validation, and deployment of **GreenSynth Analytics** — a full-stack, data-driven research and analytics platform for the green synthesis of semiconductor materials. The platform was developed through 21 sequential, iterative phases, each delivering a tested, independently verifiable increment of scientific and engineering functionality.

The core contributions of this work are:

1. A three-tier web application architecture (FastAPI + PostgreSQL + React/TypeScript) providing a scientific-grade research platform with SHA-256 enforced raw data immutability, Pydantic v2-validated API contracts, and a Docker-containerized deployment environment.
2. Four domain-specific scientific computation modules implementing the Scherrer equation for XRD crystallite size determination, the Tauc relation for UV-Vis optical band gap extraction, Ohmic resistance fitting for electrical conductivity derivation, and FTIR functional group peak annotation for CuO thin film characterization.
3. A statistically rigorous multi-sample comparison engine providing descriptive statistics, Pearson and Spearman correlation analysis, OLS regression, and one-way ANOVA with full provenance tracing to raw source data.
4. A Design of Experiments engine supporting Full Factorial, Fractional Factorial, Central Composite Design (CCD), and Box-Behnken Design with randomized run ordering, reproducible seeding, and direct conversion to planned laboratory experiments.
5. A machine learning pipeline with scientific integrity gating — including target leakage prevention, eligibility auditing, 5-fold cross-validation across five regression architectures (Baseline, Linear, Ridge, Random Forest, Gradient Boosting), applicability domain bounds enforcement, and 95% prediction confidence interval estimation. The champion Random Forest model achieved CV $R^2 = 0.6685$ and CV RMSE = 0.1501 S/cm for electrical conductivity prediction from synthesis parameters.
6. A three-level experimental prediction validation and model drift detection mechanism that closes the loop between ML predictions and laboratory outcomes, enabling scientifically honest model promotion and versioned retraining.
7. An evidence-based recommendation engine scoring and ranking synthesis parameter candidates against user-defined optimization objectives and physical domain constraints, for researcher review before conversion to planned experiments.

8. An automated scientific PDF report generation module producing multi-section, publication-quality reports with complete analytical provenance chains covering all characterization, statistical, ML, and optimization results.
9. A comprehensive test suite of 118 passing unit and integration tests (0 failures, 0 errors) verifying all scientific computation modules, API endpoints, and multi-phase pipeline workflows.

8.2 Achievements

- Successfully designed and implemented a complete, production-ready research software platform in a student academic project context, demonstrating industry-standard engineering practices including service-oriented architecture, async database access, containerization, and comprehensive automated testing.
- Achieved a 43% reduction in ML prediction error (RMSE 0.2625 $\rightarrow$ 0.1501 S/cm) through ensemble learning relative to a naive mean baseline, demonstrating the value of non-linear model architectures for capturing the physics of spray pyrolysis.
- Ingested and verified 400 experiments, 412 samples, 1,380 characterization files, and 1,881 calculated property records into a live PostgreSQL database with 100% foreign key integrity and 100% SHA-256 checksum verification across 16 atomic ingestion batches.
- Demonstrated a statistically complete DOE study generating 29 proposed CCD experiments spanning a 4-factor response surface, ready for direct laboratory execution — a capability with no equivalent in any existing open-source materials research platform.
- Demonstrated full end-to-end pipeline execution: DOE $\rightarrow$ Experiments $\rightarrow$ Characterization $\rightarrow$ Calculated Properties $\rightarrow$ Statistical Analysis $\rightarrow$ ML Training $\rightarrow$ Prediction $\rightarrow$ DOE Optimization, without breaking architectural boundaries or data integrity constraints.

8.3 Industrial usefulness

GreenSynth Analytics addresses a real and growing need in the applied materials science and green chemistry research community. The platform is directly applicable in:

- **University Research Laboratories:** As a FAIR-compliant data management and analysis platform for experimental semiconductor synthesis research, replacing fragmented spreadsheet workflows and providing traceable, reproducible analytical outputs.
- **Government Research Institutes (CSIR, DST, DRDO):** As an internally deployable analytics platform for materials characterization data management in large multi-investigator research programs.

- **Semiconductor and Solar Cell Manufacturers:** As a prototype of the kind of computational process optimization tool needed to systematically identify optimal green synthesis conditions for metal oxide thin films at pilot-plant scale, prior to full-scale manufacturing deployment.
- **Environmental Technology Companies:** For the development and optimization of CuO-based photocatalytic systems for wastewater treatment and air purification, where synthesis optimization directly impacts operational cost and treatment efficiency.

8.4 Future enhancements

The following enhancements are planned for future development phases of GreenSynth Analytics:

1. **SEM Image Segmentation:** Integration of a convolutional neural network (CNN) module for automated SEM image analysis, including grain size distribution extraction from image segmentation (Otsu thresholding, watershed algorithm) and surface morphology classification, replacing the current manual scale bar calibration approach.
2. **Multi-Objective Optimization:** Extension of the recommendation engine to support Pareto-optimal multi-objective optimization (e.g., simultaneously maximizing conductivity while minimizing crystallite size and minimizing extract concentration), using NSGA-II or similar multi-objective evolutionary algorithms.
3. **Transfer Learning for Small Datasets:** Implementation of transfer learning workflows that leverage pre-trained representations from large computational materials science databases (Materials Project, AFLOW) to improve ML model performance in the low-data regime (fewer than 50 experimental observations).
4. **Active Learning Experimental Guidance:** Integration of an active learning loop that selects the next most informative experiment from the DOE candidate space based on the current model's prediction uncertainty, enabling maximum information gain per laboratory experiment rather than passive response surface coverage.
5. **Real-Time Instrument Integration:** Development of instrument-specific data connectors (e.g., Rigaku XRD, Shimadzu UV-Vis, Bruker FTIR) for direct, automated raw data ingestion from instrument control software, eliminating the manual CSV export step.
6. **Multi-User Collaboration with Role-Based Access Control:** Full implementation of multi-user authentication, project-level access control policies (owner, collaborator, read-only reviewer), and experiment-level locking for concurrent laboratory team use.
7. **Generalization to Additional Material Systems:** Extension of the scientific computation modules to support additional metal oxide systems (ZnO, TiO₂, SnO₂, Fe₂O₃) and additional deposition techniques (Chemical Bath Deposition,

SILAR, Sol-Gel spin coating), broadening the platform applicability across the green semiconductor synthesis research community.

8. **Interactive Visualization Dashboard:** Integration of an interactive data visualization layer (using D3.js or Plotly) providing real-time Tauc plot rendering, XRD pattern overlay comparison, conductivity response surface 3D visualization, and ML prediction contour maps for the current parameter space.

Appendix A

Appendix : List of Abbreviations

ABBREVIATION	ILLUSTRATION
ANOVA	Analysis of Variance
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CCD	Central Composite Design
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
CuO	Copper Oxide
CV	Cross-Validation
DOE	Design of Experiments
DFD	Data Flow Diagram
ELN	Electronic Lab Notebook
ER	Entity-Relationship
eV	Electronvolt
FAIR	Findable, Accessible, Interoperable, Reusable
FTIR	Fourier-Transform Infrared Spectroscopy
FWHM	Full Width at Half Maximum
GB	Gradient Boosting
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
I-V	Current-Voltage
JSON	JavaScript Object Notation
JSONB	JSON Binary (PostgreSQL type)
LIMS	Laboratory Information Management System
MAE	Mean Absolute Error
ML	Machine Learning
nm	Nanometre

Appendix B

Appendix : List of Publications

The following publications are associated with this interdisciplinary mini project:

- Vaddepalli, A. N., Tolanure, J. K. S., Hede, R. M., and Bhakare, S. D., “Green-Synth Analytics: A Data-Driven Platform for Green Synthesis Optimization of CuO Thin Films,” *Proceedings of the National Conference on Emerging Trends in Computer Science and Engineering (NCETCSE)*, N. B. Navale Sinhgad College of Engineering, Solapur, 2026. (Submitted)

Appendix C

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